

A PROOF OF THE GRINBERG–LAFRENIÈRE LOWER BOUND FOR THE RANDOM-TO-BELOW SHUFFLE

ABSTRACT. For an integer $n \geq 2$ put $S(n) = \sum_{i=2}^n n / (i(H_n - H_{i-1}))$, where $H_k = \sum_{j \leq k} 1/j$. Grinberg and Lafrenière proved that $S(n)$ is the expected value of an explicit strong stationary time for the random-to-below shuffle on n cards, and conjectured that $S(n) \geq n \log n + n \log \log n$ for every $n \geq 2$ ([1, Conjecture 15.8]; $\log$ is the natural logarithm throughout). We prove this, with a uniform explicit slack:

$$S(n) \geq n \log n + n \log \log n + 0.0034553 n \quad \text{for every integer } n \geq 2.$$

The proof rests on the exact identity $S(n)/n = 1 + \log n + \log(H_n - 1) - D_n$, where D_n is a sum of convexity defects; this makes the conjecture equivalent to $D_n + \log(\log n / (H_n - 1)) \leq 1$. Bounding D_n by a telescoping estimate followed by summation by parts produces an upper bound that is non-increasing in n , so a single two-line evaluation at $n = 36$ settles every $n \geq 36$; the remaining 34 integers $2 \leq n \leq 35$ are exhibited as a table.

1. THE CONJECTURE

Grinberg and Lafrenière [1] study the *somewhere-to-below* shuffles in the symmetric group algebra. Among them is the *random-to-below* shuffle, which removes a uniformly random card and reinserts it at a uniformly random position at or below the one it came from. For an explicit strong stationary time τ of this shuffle on n cards they evaluate

$$(1) \quad \mathbb{E}(\tau) = S(n) := \sum_{i=2}^n \frac{n}{i(H_n - H_{i-1})}, \quad H_k = \sum_{j=1}^k \frac{1}{j},$$

prove the upper bound $S(n) \leq n \log n + n \log \log n + n \log 2 + 1$, and conjecture the matching lower bound:

Conjecture ([1, Conjecture 15.8]). *Let $n \geq 2$. The expected number of steps to get to the strong stationary time for the random-to-below shuffle satisfies the following lower bound:*

$$\mathbb{E}(\tau) = \sum_{i=2}^n \frac{n}{i(H_n - H_{i-1})} \geq n \log n + n \log \log n.$$

Since (1) is the authors' own evaluation of $\mathbb{E}(\tau)$ and is reprinted inside the conjecture, what remains to be proved is an inequality about the sum;

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nothing probabilistic. The hypothesis is $n \geq 2$ with no “sufficiently large n ” clause, so a proof must cover every integer.

Theorem 1. *For every integer $n \geq 2$,*

$$S(n) \geq n \log n + n \log \log n + 0.0034553 n.$$

In particular [1, Conjecture 15.8] holds.

2. NOTATION, AND AN EXACT IDENTITY

Fix $n \geq 2$ and abbreviate, for $2 \leq i \leq n$,

$$a_i := H_n - H_{i-1}, \quad b_i := \frac{1}{a_i}, \quad x_i := \frac{a_{i+1}}{a_i} \quad (2 \leq i \leq n-1).$$

Thus $a_i - a_{i+1} = 1/i$, $a_2 = H_n - 1 > 0$ and $a_n = 1/n$, and the a_i are positive and strictly decreasing, so $0 < x_i < 1$. Put

$$f(x) := x - 1 - \log x \quad (> 0 \text{ for } x \neq 1), \quad D_n := \sum_{i=2}^{n-1} f(x_i),$$

$$L_n := \log \frac{\log n}{H_n - 1}, \quad M(n) := \frac{S(n)}{n} - \log n - \log \log n.$$

All of these are defined for $n \geq 2$, because $\log n \geq \log 2 > 0$ and $H_n - 1 > 0$ there. (D_2 is an empty sum, hence 0.)

Lemma 2. *For every $n \geq 2$,*

$$(2) \quad \frac{S(n)}{n} = 1 + \log n + \log(H_n - 1) - D_n,$$

and consequently $M(n) = 1 - D_n - L_n$.

Proof. Since $a_{i+1} - a_i = -1/i$ and $a_n = 1/n$,

$$\sum_{i=2}^{n-1} (x_i - 1) = \sum_{i=2}^{n-1} \frac{a_{i+1} - a_i}{a_i} = - \sum_{i=2}^{n-1} \frac{1}{i a_i} = - \left(\frac{S(n)}{n} - \frac{1}{n a_n} \right) = 1 - \frac{S(n)}{n},$$

using $S(n)/n = \sum_{i=2}^n 1/(i a_i)$ and $1/(n a_n) = 1$. The logarithmic part telescopes:

$$- \sum_{i=2}^{n-1} \log x_i = - \log \prod_{i=2}^{n-1} \frac{a_{i+1}}{a_i} = - \log \frac{a_n}{a_2} = \log(n(H_n - 1)).$$

Adding the two displays gives $D_n = 1 - S(n)/n + \log n + \log(H_n - 1)$, which is (2). Subtracting $\log n + \log \log n$ from (2) and using $\log(H_n - 1) - \log \log n = -L_n$ gives $M(n) = 1 - D_n - L_n$. $\square$

Corollary 3. *For $c \geq 0$, the inequality $S(n) \geq n \log n + n \log \log n + cn$ is equivalent to $D_n + L_n \leq 1 - c$. The slack in the conjecture is exactly $n M(n)$.*

So the whole problem is to bound $D_n + L_n$ away from 1 from above.

3. AN UPPER BOUND FOR D_n

Lemma 4. For $n \geq 3$,

$$(3) \quad D_n \leq \frac{1}{2} \left[\frac{n}{n-1} - \frac{1}{2(H_n-1)} + R_n \right], \quad R_n := \sum_{j=3}^{n-1} \frac{1}{j(j-1)a_j}.$$

Proof. Write $x = 1 - u$ with $u \in (0, 1)$. Then $f(x) = -u - \log(1 - u) = \sum_{m \geq 2} u^m/m \leq \frac{1}{2} \sum_{m \geq 2} u^m = u^2/(2(1 - u))$, i.e.

$$(4) \quad f(x) \leq \frac{(1-x)^2}{2x} \quad (0 < x < 1).$$

Next, $1 - x_i = (a_i - a_{i+1})/a_i = (1/i)/a_i$, so

$$\frac{(1-x_i)^2}{2x_i} = \frac{1}{2i^2 a_i a_{i+1}} = \frac{1}{2i} \left(\frac{1}{a_{i+1}} - \frac{1}{a_i} \right) = \frac{1}{2i} (b_{i+1} - b_i),$$

the middle step being $a_i - a_{i+1} = 1/i$. Hence $D_n \leq \frac{1}{2} \sum_{i=2}^{n-1} \frac{1}{i} (b_{i+1} - b_i)$. Summation by parts against the decreasing weights $1/i$ gives the identity

$$\sum_{i=2}^{n-1} \frac{1}{i} (b_{i+1} - b_i) = \frac{b_n}{n-1} - \frac{b_2}{2} + \sum_{j=3}^{n-1} \frac{b_j}{j(j-1)},$$

and $b_n = n$, $b_2 = 1/(H_n - 1)$ turn this into (3). $\square$

Lemma 5. Let $N := n + 1$ and let $\lfloor \sqrt{N} \rfloor$ be the integer square root. For $n \geq 8$,

$$(5) \quad D_n \leq \Phi(n) := \frac{1}{2} \left[\frac{n}{n-1} + \frac{1}{2 \log n} + \frac{1.4426951}{\lfloor \sqrt{N} \rfloor} + \frac{4 \log(N/2)}{n-1} \right],$$

and every one of the four bracketed terms is non-increasing in n ; hence Φ is non-increasing and $D_n \leq \Phi(n_0)$ for all $n \geq n_0 \geq 8$.

Proof. We bound R_n of (3). Comparison with an integral gives

$$a_j = \sum_{k=j}^n \frac{1}{k} \geq \int_j^N \frac{dx}{x} = \log \frac{N}{j}.$$

Write $J := \sqrt{N}$ and split the range $3 \leq j \leq n-1 = N-2$ into three blocks. Since $N \geq 9$ we have $3 \leq N/2 \leq N-2$ and $J \leq N/2$, so the blocks do cover it.

Block $3 \leq j \leq J$. Here $a_j \geq \log(N/J) = \frac{1}{2} \log N$, and $\sum_{j \geq 3} 1/(j(j-1)) = 1/2$, so the block contributes at most $1/\log N$.

Block $J < j < N/2$. Here $a_j \geq \log 2$. The summation runs over the integers $j > J$, which begin at $\lfloor J \rfloor + 1$; since $\sum_{j \geq m} 1/(j(j-1)) = 1/(m-1)$, the sum $\sum_{j > J} 1/(j(j-1))$ is $1/\lfloor J \rfloor$, and so the block contributes at most $(1/\log 2)/\lfloor \sqrt{N} \rfloor \leq 1.4426951/\lfloor \sqrt{N} \rfloor$.

Block $N/2 \leq j \leq N-2$. Put $t := N-j \in [2, N/2]$. Then $a_j \geq \log(N/(N-t)) = -\log(1-t/N) \geq t/N$, and $j(j-1) \geq (N/2)((N-2)/2) = N(N-2)/4$,

so each term is at most $4/((N-2)t)$; and $\sum_{t=2}^{\lfloor N/2 \rfloor} 1/t \leq \log(N/2)$. The block contributes at most $4 \log(N/2)/(N-2)$.

Adding, $R_n \leq 1/\log N + 1.4426951/\lfloor \sqrt{N} \rfloor + 4 \log(N/2)/(N-2)$. Finally $H_n \leq 1 + \log n$ (again by integral comparison) gives $H_n - 1 \leq \log n$ and therefore $-1/(2(H_n - 1)) \leq -1/(2 \log n)$; together with $1/\log N \leq 1/\log n$ this replaces the first two terms of the bracket by $n/(n-1) + 1/(2 \log n)$, and $N-2 = n-1$ rewrites the last. That is (5).

Monotonicity. The terms $n/(n-1)$ and $1/(2 \log n)$ decrease, and $\lfloor \sqrt{N} \rfloor$ is a non-decreasing integer-valued step function of N , so its reciprocal is non-increasing. For the last term, put $\psi(N) := \log(N/2)/(N-2)$; then

$$\psi'(N) = \frac{(N-2)/N - \log(N/2)}{(N-2)^2} < 0 \quad \text{as soon as} \quad \log(N/2) > 1 - \frac{2}{N},$$

which holds for $N \geq 8$ because then $\log(N/2) \geq \log 4 > 1 > 1 - 2/N$. $\square$

4. AN UPPER BOUND FOR L_n

Lemma 6. $\gamma > 0.57721566$, where $\gamma = \lim_{n \rightarrow \infty} (H_n - \log n)$; consequently $1 - \gamma < 0.42278434$.

Proof. Let $g_n := H_n - \log n - 1/(2n)$, so $g_n \rightarrow \gamma$. With $x := 1/n > 0$,

$$g_{n+1} - g_n = \frac{1}{n+1} + \frac{1}{2n} - \frac{1}{2(n+1)} - \log\left(1 + \frac{1}{n}\right) = \frac{x(2+x)}{2(1+x)} - \log(1+x) > 0,$$

the inequality $\log(1+x) < \frac{x}{2}(1 + \frac{1}{1+x}) = \frac{x(2+x)}{2(1+x)}$ being the strict overestimation of $\int_1^{1+x} dt/t$ by the trapezoidal rule on the strictly convex integrand $1/t$. So (g_n) increases to γ , whence $\gamma > g_n$ for every n . At $n = 10^4$, exact rational arithmetic on H_{10^4} together with a rigorous enclosure of $\log 10^4$ gives

$$g_{10^4} = H_{10^4} - \log 10^4 - \frac{1}{20000} > 0.577215664068 > 0.57721566. \quad \square$$

Lemma 7. For every $n \geq 2$,

$$(6) \quad L_n \leq \Psi(n) := -\log\left(1 - \frac{0.42278434}{\log n}\right),$$

and Ψ is decreasing; hence $L_n \leq \Psi(n_0)$ for all $n \geq n_0 \geq 2$.

Proof. The sequence $H_n - \log n$ is decreasing, because $(H_{n+1} - \log(n+1)) - (H_n - \log n) = 1/(n+1) - \log(1 + 1/n) < 0$ by $\log(1+x) > x/(1+x)$. Its limit is γ , so $H_n - \log n > \gamma > 0.57721566$ by Lemma 6, i.e. $H_n - 1 > \log n - 0.42278434 > 0$ for $n \geq 2$ (note $\log 2 = 0.693 \dots > 0.4228$). Therefore

$$L_n = \log \frac{\log n}{H_n - 1} \leq \log \frac{\log n}{\log n - 0.42278434} = -\log\left(1 - \frac{0.42278434}{\log n}\right),$$

which is decreasing in n because $0.42278434/\log n$ is. $\square$

5. THE TAIL: $n \geq 36$

Proposition 8. *For every integer $n \geq 36$, $M(n) > 0.0034553$, i.e. $S(n) \geq n \log n + n \log \log n + 0.0034553 n$.*

Proof. By Lemmas 5 and 7 it suffices to check $\Phi(36) + \Psi(36) < 1 - 0.0034553$. Here $N = 37$ and $\lfloor \sqrt{37} \rfloor = 6$, so the four bracketed terms of (5) are bounded by

$$\begin{aligned} \frac{36}{35} &\leq 1.028571429, & \frac{1}{2 \log 36} &\leq 0.139527657, \\ \frac{1.4426951}{6} &\leq 0.240449184, & \frac{4 \log(18.5)}{35} &\leq 0.333459513, \end{aligned}$$

whose sum is at most 1.742007783; halving,

$$\Phi(36) \leq 0.871003892.$$

For (6), $\log 36 = 3.5835189 \dots$ and $0.42278434 / \log 36 = 0.1179802 \dots$, so

$$\Psi(36) = -\log(1 - 0.1179802 \dots) \leq 0.125540793.$$

Adding, $\Phi(36) + \Psi(36) \leq 0.996544685 < 1$, hence for all $n \geq 36$

$$M(n) = 1 - D_n - L_n \geq 1 - \Phi(36) - \Psi(36) \geq 0.003455315 > 0.0034553.$$

Corollary 3 converts this into the displayed inequality. $\square$

6. THE FINITE RANGE $2 \leq n \leq 35$

The remaining 34 integers are handled one at a time. For each, $S(n)$ is a rational number: from (1),

$$\begin{aligned} S(2) &= 2, & S(3) &= \frac{24}{5}, & S(4) &= \frac{740}{91}, & S(5) &= \frac{386080}{32571}, & S(6) &= \frac{3561064}{224257}, \\ S(7) &= \frac{915340590661}{45418769253}, & S(8) &= \frac{44929799655432}{1823748678895}, \end{aligned}$$

and so on, the numerator and denominator of $S(35)$ having 353 and 351 decimal digits respectively. Three of the rows can be read off by hand: $S(2) = 2/(2 \cdot \frac{1}{2}) = 2$ against $2 \log 2 + 2 \log \log 2 = 0.65326852 \dots$; $S(3) = \frac{9}{5} + 3 = \frac{24}{5} = 4.8$ against $3.57798035 \dots$; and $S(4) = \frac{24}{13} + \frac{16}{7} + 4 = \frac{740}{91} = 8.13186813 \dots$ against $6.85171448 \dots$

Table 1 lists, for each n , the margin $S(n) - (n \log n + n \log \log n)$, the normalised margin $M(n)$, and D_n .

Proposition 9. *For every integer n with $2 \leq n \leq 35$ one has $M(n) > 0.2674107868 > 0.0034553$, the minimum over this range being attained at $n = 7$.*

Proof. Each of the 34 values of $S(n)$ is an exact rational, computed from (1) by rational arithmetic; each right-hand side $n \log n + n \log \log n$ is enclosed in a rational interval of width below 10^{-25} . Comparing the two gives the margins of Table 1, all positive; the smallest is 1.222019651146 at $n = 3$ and the smallest normalised margin is $M(7) = 0.2674107869$. The least margin exceeds the width of the enclosures by more than twenty orders of

TABLE 1. The finite range. Columns: the margin $S(n) - (n \log n + n \log \log n)$ to 12 decimal places, $M(n)$ to 10, D_n to 9. Every margin is positive; the least is at $n = 3$, and $\min_{2 \leq n \leq 35} M(n) = M(7)$.

n	margin	$M(n)$	D_n
2	1.346731480043	0.6733657400	0.000000000
3	1.222019651146	0.4073398837	0.316290732
4	1.280153647475	0.3200384119	0.433370036
5	1.426874761492	0.2853749523	0.488200912
6	1.629643080232	0.2716071800	0.516758296
7	1.871875508071	0.2674107869	0.532388751
8	2.143632805281	0.2679541007	0.541024198
9	2.438399017356	0.2709332242	0.545623780
10	2.751643886956	0.2751643887	0.547788443
11	3.080080611049	0.2800073283	0.548438078
12	3.421244838997	0.2851037366	0.548126242
13	3.773239425541	0.2902491866	0.547198309
14	4.134571669591	0.2953265478	0.545876014
15	4.504045289409	0.3002696860	0.544304950
16	4.880686334613	0.3050428959	0.542582404
17	5.263690953326	0.3096288796	0.540774226
18	5.652387688680	0.3140215383	0.538925378
19	6.046209700325	0.3182215632	0.537066694
20	6.444673927241	0.3222336964	0.535219308
21	6.847365206445	0.3260650098	0.533397608
22	7.253923995463	0.3297238180	0.531611256
23	7.664036758584	0.3332189895	0.529866561
24	8.077428351275	0.3365595146	0.528167456
25	8.493855923618	0.3397542369	0.526516174
26	8.913103992586	0.3428116920	0.524913738
27	9.334980423760	0.3457400157	0.523360310
28	9.759313127911	0.3485468974	0.521855443
29	10.185947324781	0.3512395629	0.520398267
30	10.614743260850	0.3538247754	0.518987619
31	11.045574293407	0.3563088482	0.517622145
32	11.478325272393	0.3586976648	0.516300372
33	11.912891166026	0.3609967020	0.515020759
34	12.349175887291	0.3632110555	0.513781734
35	12.787091286981	0.3653454653	0.512581727

magnitude. Of the 34 rows only the three displayed above, $n = 2, 3, 4$, are carried out by hand, so this proposition is computer-assisted. $\square$

Proof of Theorem 1. Proposition 8 covers $\{n \geq 36\}$ and Proposition 9 covers $\{2, 3, \dots, 35\}$; the union is $\{n \geq 2\}$. Both give $M(n) > 0.0034553$, which by Corollary 3 is the assertion. $\square$

Verification. The probabilistic content is quoted, not reproved: that $S(n)$ equals $\mathbb{E}(\tau)$ for the authors’ strong stationary time is their theorem, and everything above is an inequality about the sum (1). Every numerical assertion above — the seven exact rationals and the digit counts of $S(35)$, the 34 rows of Table 1, the four displayed term bounds at $n = 36$ and the two totals, and the transcribed decimals $1.4426951 \geq 1/\log 2$, $0.57721566 < \gamma$ and $0.42278434 > 1 - \gamma$ — was recomputed from (1) in exact rational interval arithmetic with a rigorous interval logarithm, so that no comparison is decided in floating point. The program `verify.py` and its recorded output `verify.output.txt` accompany this note. Every object the program consumes is printed above, so any single row of Table 1 can also be rechecked by hand or in a computer algebra system.

REFERENCES

- [1] D. Grinberg and N. Lafrenière, *The one-sided cycle shuffles in the symmetric group algebra*, Algebraic Combinatorics **7** (2024), no. 2, 275–326. doi:10.5802/alco.346; e-print arXiv:2212.06274. Conjecture 15.8 is printed on p. 323 of the journal version and numbered Conjecture 15.14 in `arXiv:2212.06274v4`.